

BanglaMix: A Dataset and Benchmark for Code-Switching ASR in Bangladeshi Regional Dialects

M. Mohaiminul Islam

Department of Computer Science and Telecommunication Engineering
Noakhali Science and Technology University

Noakhali, Bangladesh

mohaiminul.islam@elitelab.ai

Abstract—Automatic speech recognition (ASR) for low-resource, dialectal, and code-switched speech remains an open problem. Existing Bengali resources address either regional dialects or Bengali–English code-switching (CS), but never both, even though young Bangladeshi speakers routinely mix English into regional-dialect speech. This work introduces BanglaMix, to our knowledge the first large-scale dataset for dialectal Bengali–English CS-ASR, comprising 41,499 transcribed clips (about 131 hours) spanning 15 regional dialects, collected from social-media content where natural code-switching is most common. A fully automated pipeline based on voice-activity segmentation, Whisper-based silver-standard transcription, script-based code-switch detection, and a single large language model (LLM) judge for borderline clips removes the need for manual transcribers. Three ASR families are benchmarked under zero-shot and fine-tuned conditions: Whisper, Meta MMS, and wav2vec2-XLS-R. A controlled ablation over training subsets (dialect-only, code-switch-only, and combined) isolates the contribution of each data type. The fine-tuned Whisper-small model reaches a word error rate (WER) of 0.80 and a character error rate (CER) of 0.49, a 25% relative WER reduction over its zero-shot counterpart and a statistically significant improvement ($p < 0.001$, paired bootstrap) over every zero-shot baseline and a fine-tuned CTC model, including zero-shot Whisper large-v3. Absolute error rates remain high because dialectal Bengali has no standardized orthography and the labels are silver-standard; a human validation of 61 clips quantifies the label quality at WER 0.70 / CER 0.39, and BanglaMix is therefore positioned as a challenging, open benchmark for the community. The dataset and full construction-and-evaluation code are publicly available¹ to support future research.

Index Terms—Automatic speech recognition, Bengali ASR, code-switching, dialectal speech, low-resource speech recognition, Whisper, large language model judge.

I. INTRODUCTION

Speech is the most natural interface to technology, yet automatic speech recognition (ASR) quality is highly uneven across languages and speaking styles. Two factors that strongly degrade ASR are *dialectal variation* and *code-switching* (CS), the practice of mixing words from a second language into an utterance. Both are pervasive in everyday Bangladeshi speech:

the country has many mutually distinct regional dialects, and younger, educated speakers freely insert English words into dialectal Bengali, producing utterances such as “*chatgaiya bhashay phone unboxing kortachi, honestly speaking camera performance onek bhalo*” (“unboxing a phone in the Chittagong dialect; honestly speaking, the camera performance is very good”).

Despite this, existing Bengali speech resources cover only one phenomenon at a time. Read-speech corpora such as OpenSLR-37 [1] target standard Bengali; the Ben-10 corpus [2] covers ten dialects but contains no code-switching; and the MUCS shared task [3] provides Bengali–English CS only in standard (Dhaka) Bengali. No dataset or model addresses their *intersection*—dialectal Bengali speech mixed with English—which is precisely the dominant informal communication style.

Recognizing this kind of speech is hard for several compounding reasons, and understanding them motivates the design choices made throughout this work.

- **Acoustic divergence.** Regional dialects differ from standard Bengali in vowel quality, consonant clusters, prosody, and vocabulary, so acoustic models trained on standard Bengali generalize poorly.
- **Two scripts in one utterance.** Code-switching forces the model to move between Bengali and Latin scripts within a single utterance, which monolingual recognizers cannot represent at all.
- **Data scarcity.** Dialectal and code-switched speech is rare in existing corpora, and paid transcription of 15 dialects is prohibitively expensive for a low-resource language.
- **No standardized orthography.** Dialect words and borrowed English words have no single accepted spelling, so even a perfect transcription can be “wrong” under a strict word-level metric.

These four factors shape the dataset pipeline (which avoids manual transcription and uses script-based code-switch detection), the model choice (which favors a shared-vocabulary sequence-to-sequence model), and the evaluation (which emphasizes character-level error and code-switch-specific met-

¹Dataset: https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset; code: <https://github.com/niloycste/ASR-Bangali-Local-Dialect->

rics).

This work fills that gap; Fig. 1 gives an end-to-end overview of the dataset construction, models, and evaluation. The main contributions are as follows.

- 1) **BanglaMix dataset.** To our knowledge, the first large-scale dataset for dialectal Bengali–English CS-ASR: 41,499 transcribed clips (about 131 hours) across 15 regional dialects, sourced from social media using dialect-name search queries.
- 2) **Automated, annotator-free pipeline.** An end-to-end pipeline using voice-activity segmentation, Whisper-based silver-standard transcription, script-based CS detection, and a single strong multilingual LLM judge for borderline transcripts, enabling large-scale annotation without manual transcribers.
- 3) **Systematic benchmark and ablation.** A controlled comparison of three ASR families under zero-shot and fine-tuned conditions, with a training subset ablation (dialect-only, CS-only, combined) that isolates the value of each data type.
- 4) **Honest analysis.** A detailed error, per-dialect, and code-switching analysis, with statistical significance testing, that characterizes *where* current models fail and establishes BanglaMix as a hard open benchmark.

We stress that the high absolute error rates reported here are the central *finding* of this work, not a shortcoming of it: that a model trained on 680,000 hours of audio still attains only 0.91 WER on this speech is direct evidence that dialectal code-switched Bengali lies almost entirely outside the reach of current ASR, leaving its many millions of speakers unserved. The contribution is therefore the *resource and the measurement*—the first dataset that makes this gap quantifiable, together with an honestly characterized silver-label noise floor (Section III-E)—rather than a state-of-the-art accuracy number. A resource that breaks the strongest available systems is precisely what is needed to direct future effort toward this underserved population.

The remainder of this paper is organized as follows. Section II reviews related work on Bengali, dialectal, and code-switching ASR and positions this work against it. Section III details the BanglaMix dataset and its construction pipeline. Section IV describes the ASR models and the fine-tuning procedure, and Section V presents the experimental setup. Section VI reports the main results, and Section VII provides an in-depth error, code-switching, ablation, and case-study analysis. Section VIII discusses limitations and reproducibility, and Section X concludes with directions for future work. Table I summarizes the notation used throughout.

II. RELATED WORK

A. Bengali and Dialectal ASR

Early Bengali ASR relied on Gaussian-mixture and deep-neural-network hidden Markov models with hand-built pronunciation lexicons [4], [5], which were limited to standard Bengali and struggled with out-of-vocabulary words.

TABLE I
SUMMARY OF NOTATION.

Symbol	Description
η_{lo}, η_{hi}	low/high no-speech acceptance thresholds (0.30, 0.60)
n_{bn}, n_{en}	number of Bengali / English words in a clip
b	number of intra-utterance language boundaries
ρ	English-word ratio $n_{en}/(n_{en} + n_{bn})$
S, D, I	substitutions, deletions, insertions in the alignment
N	number of reference tokens
P, R	precision and recall of code-switch detection
$\mathcal{B}$	CTC alignment-collapsing function

The field then moved to end-to-end models: connectionist temporal classification (CTC) [6], deep recurrent networks [7], Deep Speech [8], and attention-based encoder-decoders such as Listen, Attend and Spell [9], supported by toolkits such as ESPnet [10], which lowered error rates on standard Bengali [11]. The Transformer [12] and masked pre-training [13] subsequently enabled large self-supervised and weakly-supervised speech models—wav2vec 2.0 [14], HuBERT [15], WavLM [16], the multilingual XLS-R [17], [18], Whisper [19], and MMS [20]—which, together with multilingual corpora such as Common Voice [21], improved transfer to low-resource Bengali. Dialectal Bengali received focused attention only recently. The Ben-10 corpus [2] covers ten regional dialects (over 78 hours, 394 speakers) and shows that even strong foundation models, including Whisper large-v3, perform poorly on out-of-distribution dialect speech, both zero-shot and after fine-tuning. Biswas *et al.* [22] propose a WavLM-based denoising and adaptation framework for dialectal Bengali (e.g., Chittagonian and Sylheti) under simulated noise, reporting large relative WER reductions over wav2vec 2.0 and Whisper at low signal-to-noise ratios. Ridoy *et al.* [23] compare Whisper and Wav2Vec-BERT on standard Bangla and find the self-supervised encoder both more accurate and more compute-efficient under low-resource conditions. None of these, however, addresses code-switching, which motivates a benchmark that combines dialect and code-switching in a single resource.

B. Code-Switching ASR

Code-switching—alternating between two languages within a conversation or even a single utterance—is pervasive in multilingual societies and especially common in informal Bangladeshi speech, where English words are routinely embedded in Bengali sentences. It is hard for ASR for several compounding reasons: the effective vocabulary spans two languages and, for Bengali–English, two scripts; code-switched utterances are rare in standard training corpora; and language switches interact with accent and acoustic variation, complicating both the acoustic and language-modelling components [24], [25]. Dedicated corpora have driven much of the progress in the field, most prominently the Mandarin–English SEAME corpus [26], while recent analysis shows that even large multilingual models handle code-switching unreliably

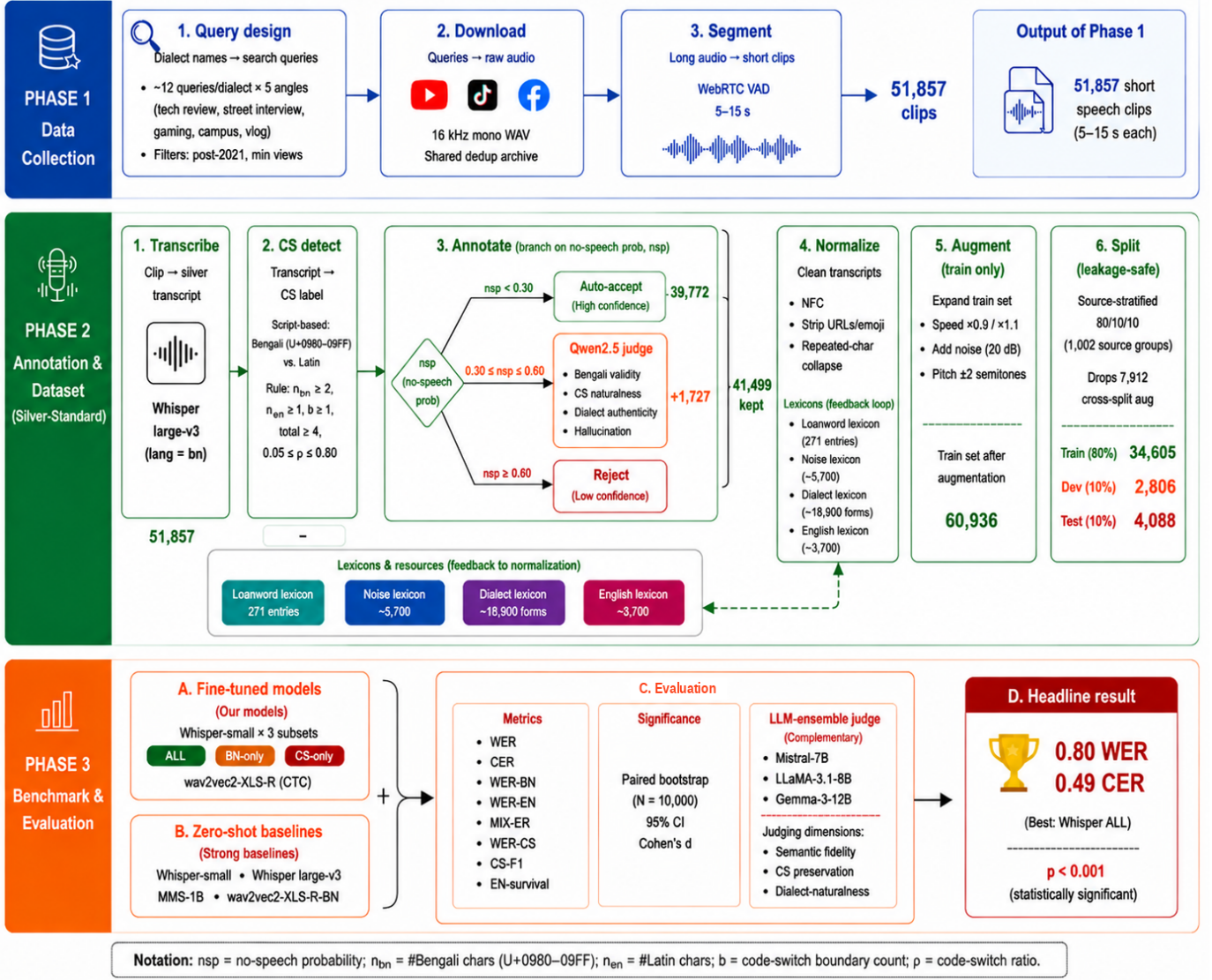


Fig. 1. End-to-end overview of BanglaMix. Dialect-targeted social-media collection and voice-activity segmentation feed a fully automated silver-standard labeling stage (Whisper large-v3 transcription, script-based code-switch detection, and an LLM judge), yielding 41,499 transcribed clips. After normalization and a leakage-safe, source-stratified split, fine-tuned and zero-shot ASR models are benchmarked with edit-distance, code-switching, and LLM-ensemble metrics.

unless given explicit code-switched supervision [27]. For Indian languages, the MUCS 2021 challenge [3] introduced Bengali–English and Hindi–English code-switching benchmarks, but its Bengali portion is restricted to standard (Dhaka) Bengali and contains no regional dialects. A useful property of the Bengali–English pair is that the two languages occupy disjoint Unicode ranges, so word-level language identification is almost exact from script alone, without a dedicated language-identification model. Architecturally, sequence-to-sequence models such as Whisper are well suited to code-switching because their shared sub-word vocabulary [28] already covers both scripts and can emit mixed-script text in a single decoding pass, whereas monolingual CTC models such as MMS must commit to one output script and cannot

represent code-switching at all. This work evaluates both paradigms in order to quantify that gap on dialectal Bengali.

C. Automated Corpus Construction for Low-Resource ASR

Because manual transcription does not scale to many low-resource varieties, recent corpora are built with automated pipelines. GigaSpeech 2 [29] assembles a large multi-domain corpus for Thai, Indonesian, and Vietnamese through automated web crawling, Whisper-based transcription, and iterative refinement, demonstrating that weakly supervised labels can replace human transcribers at scale. Complementary work uses synthetic speech and spectrogram augmentation to enlarge low-resource training pools, including for Bengali [30]. BanglaMix follows this automated philosophy but targets a

previously uncovered setting—multi-dialect Bengali with English code-switching—and adds an LLM judge for borderline transcripts.

D. LLM-as-a-Judge for Data Quality

Using large language models (LLMs) as automatic evaluators has become common in natural-language processing, ranging from pairwise response judging [31], [32] to multi-dimensional quality scoring [33]. Such judges are attractive precisely when human annotation is costly and reference labels are scarce or noisy—as in low-resource dialectal ASR—although their reliability depends on prompt design and the capacity of the judging model. In this work an LLM judge is used at two distinct stages. During *dataset construction*, a single strong multilingual model (Qwen2.5 [34]) scores borderline silver-standard transcripts along several quality dimensions and decides whether to retain them. During *evaluation*, an ensemble of three open models—Mistral [35], LLaMA [36], and Gemma [37]—provides an auxiliary, reference-free signal that complements word-level metrics (Section VII). Applying LLM judges to both quality control and evaluation for code-switched, low-resource ASR has, to the authors’ knowledge, not been explored before.

E. Positioning of This Work

Table II summarizes representative Bengali and code-switching ASR resources along the dimensions most relevant to this work. Two gaps are clear. Standard-Bengali corpora and models [1], [4], [5], [11] address neither dialect nor code-switching. The dialectal corpus Ben-10 [2] covers multiple dialects but no code-switching, while the MUCS task [3] covers code-switching but only in standard Bengali. General multilingual models [17], [19], [20] transfer to standard Bengali but degrade sharply on dialects and, for monolingual CTC models, cannot emit code-switched output at all. BanglaMix is, to our knowledge, the first resource to occupy the intersection: *multi-dialect Bengali with natural English code-switching*.

III. THE BANGLAMIX DATASET

A. Design Principles and Why They Matter

BanglaMix is built around three principles. First, *dialectal authenticity*: the audio must contain genuine regional dialect, not standard Bengali spoken by someone from a region. This matters because region-name queries (for example, “Chit-tagong vlog”) mostly retrieve people *from* a region speaking standard Bengali to a general audience; dialect-name queries instead surface content where creators deliberately use the dialect. Second, *natural code-switching*: CS must arise spontaneously in informal speech rather than from scripted content, because scripted CS does not reflect real acoustic and lexical patterns. Third, *scalability*: the pipeline must be fully automated so the dataset can grow without paid transcribers, which is essential for a low-resource language where annotation budgets are small.

Algorithm 1 BanglaMix: From Raw Dialectal Audio to a Curated CS-ASR Benchmark

```

1: Input: dialect-name queries  $\mathcal{Q}$ ; VAD aggressiveness  $a$ ; no-
   speech thresholds  $\tau_o=0.30$ ,  $\tau_{hi}=0.60$ ; split ratio 80/10/10
2: Output: curated dataset  $\mathcal{D} = (\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te})$ 
   Step 1: Acquire and segment
3:  $\mathcal{A} \leftarrow \text{DOWNLOAD16K}(\mathcal{Q})$   $\triangleright$  yt-dlp + ffmpeg; shared dedup
   archive
4:  $\mathcal{C} \leftarrow \text{VADSEGMENT}(\mathcal{A}, a)$   $\triangleright$  WebRTC VAD; 5–15 s clips  $\Rightarrow$ 
   51,857
   Step 2: Transcribe and detect code-switching
5: for each clip  $c \in \mathcal{C}$  do
6:    $t_c \leftarrow \text{WHISPERLARGEV3}(c, \text{lang=bn})$ 
7:    $c.\text{CS} \leftarrow \text{SCRIPTCSDetect}(t_c)$   $\triangleright$  Bengali vs. Latin
   Unicode ranges
8: end for
   Step 3: Silver-standard annotation (no human transcribers)
9:  $\mathcal{A} \leftarrow \{c : \text{NSP}(c) < \tau_o\}$   $\triangleright$  auto-accept  $\Rightarrow$  39,772
10:  $\mathcal{B} \leftarrow \{c : \tau_o \leq \text{NSP}(c) < \tau_{hi}\}$   $\triangleright$  borderline  $\Rightarrow$  8,712
11: for each  $c \in \mathcal{B}$  do
12:    $d \leftarrow \text{QWEN2.5JUDGE}(t_c)$   $\triangleright$  4 dimensions;
   accept/reject/uncertain
13:   if  $d = \text{ACCEPT}$  then
14:      $\mathcal{A} \leftarrow \mathcal{A} \cup \{c\}$   $\triangleright +1,727$ 
15:   end if
16: end for
   Step 4: Normalise transcripts
17:  $\forall c \in \mathcal{A} : t_c \leftarrow \text{NORMALISE}(t_c)$ 
   Step 5: Augment (training pool only)
18:  $\mathcal{G} \leftarrow \text{AUGMENT}(\mathcal{A})$ 
   Step 6: Quality filter and leakage-safe split
19:  $\mathcal{A} \leftarrow \text{DROPLowQUALITY}(\mathcal{A} \cup \mathcal{G})$ 
20:  $(\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te}) \leftarrow \text{SOURCESTRATSplit}(\mathcal{A}, 80/10/10)$ 
21:  $\mathcal{D}_{tr} \leftarrow \mathcal{D}_{tr} \cup \text{PINAUGToTRAIN}(\mathcal{G})$ 
22: return  $\mathcal{D}$   $\triangleright$  final curated dataset

```

B. Query Design

Search queries are the single most important lever for collecting authentic dialectal code-switched speech, because they determine *which* creators are retrieved. As noted above, region-name queries surface standard-Bengali content, so this work instead uses *dialect-name* queries combined with content genres in which spontaneous code-switching is most frequent. For each dialect, around a dozen queries are designed across five content angles, summarized in Table III. The rationale is that street-interview and live-reaction content is spontaneous and cannot sustain a fake accent, technology and campus content elicits English technical terms, and vlog content yields relaxed, mixed informal speech. Quality filters exclude pre-2021 uploads and videos with negligible viewership.

C. Construction Pipeline

Algorithm 1 summarizes the end-to-end pipeline; each step is justified below.

Step 1: acquisition and segmentation: Audio is downloaded at 16 kHz mono using yt-dlp [38] from social-media platforms (YouTube, TikTok, and Facebook), with a shared de-duplication archive so the same video is never collected twice. Long recordings are segmented into 5–15 s clips using WebRTC voice-activity detection [39]. Segmentation is necessary because raw social-media audio contains long non-speech

TABLE II

COMPARATIVE SUMMARY OF REPRESENTATIVE BENGALI AND CODE-SWITCHING ASR RESOURCES AND MODELS. “CS” INDICATES CODE-SWITCHING SUPPORT; “–” DENOTES NOT APPLICABLE. BANGLAMIX IS THE FIRST TO COMBINE MULTI-DIALECT BENGALI WITH ENGLISH CODE-SWITCHING.

Work	Method / Model	Objective	Variety	CS	Hours	Key limitation
Hasnat et al. [4]	GMM-HMM	Isolated-word Bengali ASR	Standard BN	No	–	Lexicon-dependent; WER >50%; no OOV handling
Swarna et al. [5]	Survey (HMM/GMM/DNN)	Bengali phoneme recognition survey	Standard BN	No	–	Survey; standard Bengali, no dialect or CS
Mandal et al. [11]	End-to-end (CTC/attention)	Lexicon-free Bengali ASR	Standard BN	No	–	Standard Bengali only; no dialect or CS
Kjartansson et al. [1]	OpenSLR-37 corpus	Read-speech benchmark	Standard BN	No	100	Read speech; no dialect, no code-switching
MUCS 2021 [3]	CS challenge corpus	Bengali-English CS-ASR	Standard BN	Yes	16	Standard Dhaka Bengali; no dialect
Ben-10 [2]	Whisper fine-tuning	Multi-dialect Bengali ASR	Dialectal BN	No	78	No code-switching
XLS-R [17]	wav2vec2 self-supervised	Multilingual transfer	Multilingual	No	–	Standard Bengali via transfer; no dialect/CS
Whisper [19]	Weakly-supervised seq2seq	Multilingual ASR	Multilingual	No	–	~70% WER on unseen Bengali dialects
MMS [20]	CTC with adapters	1000+ language ASR	Multilingual	No	–	Monolingual output; cannot emit code-switching
Ridoy et al. [23]	Whisper vs. Wav2Vec-BERT	Low-resource Bangla model comparison	Standard BN	No	~90–100	Standard Bangla only; no dialect or CS
Biswas et al. [22]	WavLM denoising + adaptation	Noise-robust dialectal Bengali ASR	Dialectal BN	No	~20	Dialectal but no code-switching; noise-focused
GigaSpeech 2 [29]	Automated crawl + transcribe + refine	Large-scale low-resource corpus	Th/Id/Vi	No	~30k	Not Bengali; no dialect or code-switching
BanglaMix (this work)	(this Dataset Whisper/CTC benchmark)	+ Dialectal Bengali-English CS-ASR	Dialectal BN	Yes	131	First multi-dialect Bengali with code-switching

TABLE III

CONTENT ANGLES USED TO DESIGN DIALECT-NAME SEARCH QUERIES, AND WHY EACH ELICITS AUTHENTIC DIALECTAL CODE-SWITCHING.

Content angle	Why it elicits dialectal CS
Technology / phone review	Forces English technical terms within dialect speech
Street interview / reaction	Spontaneous public responses; a fake accent cannot be sustained
Gaming / live stream	Live commentary with frequent English game terms
Campus / career	Academic and job English mixed into regional speech
Vlog / content creator	Relaxed informal speech with frequent CS

regions, and fixed 5–15s clips match the input length that modern ASR models train on, which keeps both transcription and fine-tuning efficient. This step yields 51,857 clips.

Step 2: transcription and CS detection: Each clip is transcribed with Whisper large-v3 [19] with the language forced to Bengali. Whisper large-v3 is chosen as the labeler because it is the strongest openly available multilingual model and therefore the best available proxy for a human transcriber on this hard data. Code-switching is detected directly from Unicode script ranges: Bengali (U+0980–U+09FF) and Latin (ASCII) occupy

disjoint ranges, so word-level language identification is nearly exact without a dedicated model. Let n_{bn} and n_{en} be the number of Bengali- and Latin-script words in a clip, b the number of language boundaries (adjacent words of different scripts), and $\rho = n_{en}/(n_{en} + n_{bn})$ the English-word ratio. A clip is labeled code-switched when

$$(n_{bn} + n_{en}) \geq 4 \wedge n_{bn} \geq 2 \wedge n_{en} \geq 1 \wedge b \geq 1 \wedge 0.05 \leq \rho \leq 0.80. \quad (1)$$

A stricter high-confidence subset additionally requires $n_{en} \geq 2$ and $0.10 \leq \rho \leq 0.70$. The lower bound on ρ excludes isolated borrowed words and the upper bound excludes English-dominant clips, so that only genuinely mixed utterances are tagged; the same rule is applied to model hypotheses when computing code-switch detection F1.

Step 3–4: silver-standard annotation with an LLM judge: Manual transcription of dialectal CS speech at scale is infeasible, so a two-stage silver-standard strategy is used. High-confidence transcripts (Whisper no-speech probability below 0.30) are accepted directly, following evidence that high-confidence Whisper output is a usable training signal [29], [40]; clips above 0.60 are rejected as noise. The uncertain middle band is the hard case, and here a single strong multilingual judge (Qwen2.5 [34]) scores each transcript on Bengali validity, code-switch naturalness, dialect authenticity, and absence of hallucination, and returns accept/reject/uncertain. A

single high-capacity judge is used at this stage—rather than an ensemble—because the borderline set is large and a single strong model offers the best quality-per-compute trade-off; running it locally also avoids API cost and keeps the data private. After this stage, 41,499 clips remain.

Step 5: normalization: Transcripts are cleaned with Unicode NFC normalization, removal of zero-width characters, URLs, hashtags, mentions and emoji, digit normalization, repeated character collapse, and punctuation/whitespace cleanup. A second, evaluation-only step maps spelling variants of borrowed English words to a single canonical form using a curated lexicon of 272 entries (650 variant spellings). This is important because dialectal Bengali has *no standardized orthography* for loanwords, so the same spoken word may be written several valid ways; without this mapping, word error rate would be inflated by legitimate spelling variation rather than genuine recognition errors. The pipeline additionally maintains three supporting lexicons that encode dialect- and code-switch-specific knowledge: a list of about 5,700 known ASR hallucination and noise tokens, used to detect and remove corrupted transcripts; a list of about 18,900 valid dialectal Bengali word forms that are explicitly *protected* from being normalized toward standard spellings, so genuine dialect vocabulary is preserved; and a list of about 3,700 common English words that supports the script-based identification of code-switched tokens. These lexicons make the otherwise generic normalization sensitive to the two phenomena—regional dialect and code-switching—that this dataset targets.

Step 6–7: augmentation, quality filter, and leakage-safe split: Audio augmentation (speed perturbation, additive noise, pitch shift) is applied to the training pool to improve robustness to acoustic variation, complementing spectrogram-level methods such as SpecAugment [41]. The data is then split at the *source* level—all clips from one original video stay in one split—to prevent speaker and channel leakage, which would otherwise inflate test scores. The split is dialect-stratified so every dialect appears in all three partitions. Augmented clips inherit their source and are pinned to training only when that source is in the training split, so augmentation never leaks a source across splits.

D. Statistics

Table IV lists the 15 dialects and Fig. 2 shows the clip distribution, which is intentionally broad but naturally imbalanced, mirroring online content availability. Of the 51,857 segmented clips (≈ 164 hours of raw audio), 41,499 (131.2 hours) pass silver-standard annotation and constitute the released dataset—the remaining ≈ 33 hours are clips rejected by the no-speech threshold or the LLM judge; Table V reports its train/dev/test splits. The dataset’s primary scale is its dialect breadth—15 varieties over 131 hours—while code-switching forms a deliberate *natural-prevalence* minority (5.7% of clips): English insertion is a spontaneous, in-the-wild phenomenon, and we capture it at its real frequency rather than artificially over-sampling it. Code-switching is therefore treated as a distinct evaluation axis layered on top of broad dialect coverage, and

TABLE IV
DIALECT COVERAGE IN THE RELEASED DATASET.

Dialect	Clips	Hours
Sylhet	7,931	26.0
Old Dhaka	4,741	14.6
Sandwip	4,583	13.0
Barishal	3,259	11.0
Chittagong	2,993	9.1
Khulna	2,774	8.9
Narail	2,371	7.8
Kishoreganj	2,287	7.1
Rangpur	1,985	6.4
Comilla	1,847	6.1
Narsingdi	1,661	5.1
Habiganj	1,446	4.5
Mymensingh	1,369	4.1
Noakhali	1,140	3.6
Tangail	1,112	3.8
Total	41,499	131.2

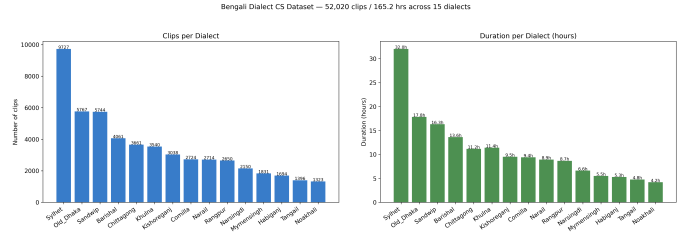


Fig. 2. Clip distribution across the 15 dialects.

CS-specific results (Section VI) are reported on this naturally small subset accordingly.

E. Human Validation of Silver-Standard Labels

Because the transcripts are silver-standard, we quantify their reliability against human reference. A bilingual annotator transcribed 61 randomly drawn test clips (2–6 per dialect, balanced across the code-switching axis) by listening to the audio; the human transcript is treated as the reference and the silver label as the hypothesis. Overall agreement is **WER** 0.70 and **CER** 0.39. The large WER–CER gap is informative: at the character level the labels are far closer to human transcription than the word-level figure suggests, reflecting the absence of a standardized orthography for these dialects—annotators and Whisper frequently choose different spellings of the same spoken word, which inflates WER without indicating a semantic error.

Two patterns are consistent across the sample. First, code-switched labels are *more* reliable than Bengali-only ones (WER 0.58 vs. 0.74): the embedded English tokens act as clear acoustic anchors that the transcription model captures well. Second, label quality varies by dialect (Table VI): Sandwip, Kishoreganj, Sylhet, Khulna, and Old Dhaka are transcribed reliably (WER ≤ 0.54 , CER ≤ 0.34), whereas Habiganj and Noakhali are weakest (WER > 1.0 , CER ≈ 0.6), driven by

TABLE V
TRAIN/DEV/TEST SPLITS OF THE RELEASED (REAL, UNAUGMENTED) DATASET. AUDIO AUGMENTATION EXPANDS THE TRAINING SPLIT TO 60,936 CLIPS FOR MODEL TRAINING; DEV AND TEST REMAIN REAL AND UNAUGMENTED.

Split	Clips	Hours	CS clips	CS%
Train	34,605	108.9	2,019	5.8
Dev	2,806	9.1	166	5.9
Test	4,088	13.2	176	4.3
Total	41,499	131.2	2,361	5.7

TABLE VI
HUMAN VALIDATION OF SILVER-STANDARD LABELS ON 61 TEST CLIPS (HUMAN REFERENCE VS. SILVER HYPOTHESIS). LOWER IS BETTER; n IS CLIPS PER DIALECT.

Dialect	n	WER	CER
Sandwip	3	0.26	0.10
Kishoreganj	5	0.46	0.25
Khulna	3	0.53	0.32
Old Dhaka	6	0.54	0.31
Sylhet	4	0.54	0.34
Narail	3	0.63	0.27
Barishal	5	0.72	0.39
Tangail	4	0.74	0.41
Rangpur	6	0.75	0.51
Narsingdi	3	0.76	0.38
Chittagong	4	0.80	0.40
Mymensingh	6	0.84	0.47
Comilla	2	0.92	0.44
Noakhali	3	1.04	0.59
Habiganj	4	1.06	0.60
Overall	61	0.70	0.39

Whisper insertions and hallucination on their more divergent phonology. These figures bound the silver-label noise floor, contextualize the absolute error rates reported in Section VI, and motivate label-refinement directions (e.g. judge-guided self-training) for future work. Per-dialect estimates rest on small samples ($n = 2-6$) and are therefore indicative rather than definitive.

F. Ethical and Legal Considerations

All audio is sourced from publicly available social-media content. To respect creators and platform terms, the released dataset will distribute segmented audio clips and transcripts for non-commercial research use, together with source identifiers and timestamps so that the original content can be located and attribution preserved. Because the data is conversational and may contain personal opinions, no speaker identities, demographic labels, or contact information are collected or released, and the dataset is intended only for building and evaluating speech-recognition systems. Dialect coverage follows online content availability rather than population, so it is intentionally not a demographic sample; this imbalance is reported transparently (Fig. 2) so that downstream users can weight or resample as needed.

Several further considerations apply. *Consent and platform terms*: the source content is public, but its creators did not explicitly consent to research use; we therefore release the data strictly for non-commercial research, honor platform Terms of Service and copyright by distributing short derived segments with source identifiers rather than re-hosting whole videos, and we provide a contact-based *takedown/opt-out* process so any creator can request removal of their content. *Voice as personal data*: speech is biometric and potentially identifying, so we release no speaker names, handles, faces, or contact information, segment audio into short clips that hinder reconstruction of full recordings, and explicitly prohibit speaker-identification, surveillance, or re-identification uses in the accompanying license. *Potential harms*: as with any ASR resource, downstream models could be misused or could underperform for less-represented dialects; we mitigate this through per-dialect reporting, transparent disclosure of label noise (Section III-E), and an intended-use statement limiting the dataset to building and evaluating ASR for these under-served varieties. A formal ethics-board review was not required for the use of public, de-identified media under the host institution’s policy, but the above safeguards were applied throughout.

IV. ASR MODELS AND FINE-TUNING

This work evaluates three model families, chosen to cover the two dominant ASR paradigms and a range of pre-training regimes.

Whisper (sequence-to-sequence): Whisper [19] is an encoder-decoder transformer trained on 680k hours of weakly supervised multilingual audio. It is central to this work because its sub-word vocabulary contains both Bengali and Latin tokens, so it can emit mixed-script (code-switched) output in a single decoding pass—a natural fit for the task. While convolution-augmented architectures such as the Conformer [42] are also competitive for ASR, Whisper’s shared multilingual vocabulary makes it especially suited to code-switching. The fine-tuned models use Whisper-small (244M parameters), which is feasible to train on modest hardware while still benefiting from large-scale pre-training. All inference forces Bengali transcription.

Meta MMS (CTC, multilingual): MMS [20] is a CTC model covering 1,000+ languages with per-language adapters. It is included as a strong multilingual zero-shot baseline; because its output vocabulary is monolingual Bengali, it cannot emit English script, which makes it a useful contrast for measuring how much code-switching specifically hurts.

wav2vec2-XLS-R (CTC, self-supervised): wav2vec2-XLS-R-300M [17] is pre-trained on 436k hours of multilingual audio. A Bengali-fine-tuned checkpoint is used as a zero-shot baseline, and it is additionally fine-tuned on BanglaMix to represent the CTC paradigm under the same training data as Whisper.

Fine-tuning procedure: Whisper is fine-tuned with the Hugging Face Transformers [43] and Datasets [44] libraries using the Adam optimizer [45]; learning rate 10^{-5} , effective batch size 16, three epochs, 500 warm-up steps, with the

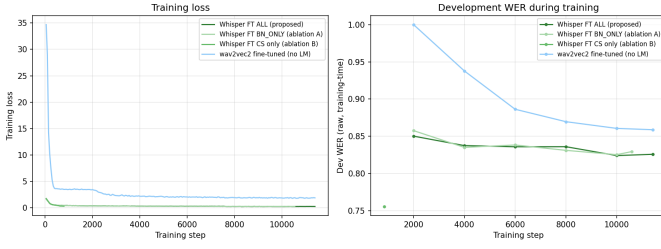


Fig. 3. Training loss (left) and development WER (right) during fine-tuning. The development WER is the raw training-time value and is higher than the final normalized test WER in Section VI.

TABLE VII
FINE-TUNING HYPERPARAMETERS FOR THE TWO TRAINED MODEL FAMILIES.

Hyperparameter	Whisper-small	wav2vec2-XLS-R
Base parameters	244M	300M
Architecture	seq2seq	CTC
Learning rate	10^{-5}	10^{-4}
Effective batch size	16	16
Epochs	3	3
Warm-up steps	500	500
Precision	bf16/fp16	bf16/fp16
Model selection	best dev WER	best dev WER

best checkpoint selected by development WER. For the CTC model, a character vocabulary covering Bengali, Latin, and common punctuation is built from the training transcripts, the CTC head is reinitialized, and the encoder is fine-tuned at learning rate 10^{-4} for three epochs. All fine-tuning and benchmark inference (zero-shot baselines and fine-tuned systems) run on a single NVIDIA A100-SXM4-40GB GPU, whereas the one-time silver-standard transcription of all clips with Whisper large-v3 was performed on a CPU workstation (Intel Core i9-14900, 24 cores, 32 GB RAM). The CTC model is trained to minimize the negative log-likelihood marginalized over all alignments π that collapse to the target y ,

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_t p(\pi_t | x), \quad (2)$$

where $\mathcal{B}$ is the CTC collapsing function, whereas Whisper is trained with the standard token-level cross-entropy of its decoder. Key hyperparameters are listed in Table VII. Fig. 3 shows that all Whisper variants converge within about 2k steps; the CS-only ablation runs for fewer steps because it uses only the roughly 4.5k code-switched clips.

Training subset ablation: To isolate the contribution of each data type, three Whisper variants are trained: **BN_ONLY** (dialect-only clips), **CS_ONLY** (code-switched clips only), and **ALL** (both). All three share the same base model, procedure, and test set, so any difference is attributable to the training data composition.

V. EXPERIMENTAL SETUP

A. Metrics and Why Each is Used

Word error rate (WER) and character error rate (CER) are reported. CER is given prominence because, with no standardized dialectal orthography, word-level WER over-penalizes valid spelling variants, whereas CER better reflects how well the sounds are captured. WER is further decomposed into WER-BN and WER-EN (errors on Bengali and English words) to localize failures, and a mixed-error rate (MIX-ER) measures combined script handling. Code-switch detection F1 (CS-F1) measures whether a model produces any code-switching at all. All references and hypotheses pass the same normalization (Section III), including collapse of degenerate repetition, before scoring, using standard error-rate implementations [46], [47].

Formally, with S , D , and I the number of substitutions, deletions, and insertions from the reference-hypothesis alignment and N reference tokens,

$$\text{WER} = \frac{S + D + I}{N}, \quad (3)$$

computed at the word level. CER applies the same edit-distance formula at the character level,

$$\text{CER} = \frac{S_c + D_c + I_c}{N_c}, \quad (4)$$

where the subscript c denotes character counts; CER is more robust to inconsistent dialectal spelling and is therefore the primary metric in this work. WER-BN and WER-EN restrict the alignment to Bengali-script and Latin-script reference words, respectively, while the mixed error rate (MIX-ER) treats each token as a (word, script) pair and counts a token correct only when both the word *and* its language match, jointly penalizing recognition and script errors. Code-switch detection treats “does the hypothesis contain code-switching” as a binary decision; with TP, FP, and FN the true positives, false positives, and false negatives against the script-based CS label,

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{CS-F1} = \frac{2P \cdot R}{P + R}. \quad (5)$$

Finally, English-word survival quantifies code-switch fidelity as the fraction of Latin-script reference words on code-switched clips that also appear in the hypothesis,

$$\text{EN-Surv} = \frac{\sum_{(r,h) \in \mathcal{C}} |\{w \in r : \text{lat}(w)\} \cap h|}{\sum_{(r,h) \in \mathcal{C}} |\{w \in r : \text{lat}(w)\}|}, \quad (6)$$

where $\mathcal{C}$ is the set of code-switched reference-hypothesis pairs and $\text{lat}(w)$ is true for Latin-script words.

B. Baselines

Four zero-shot baselines are used: *Whisper-small*, the same base model as the proposed system evaluated untuned, which isolates the effect of fine-tuning from model scale; *Whisper large-v3*, the strongest openly available model; *MMS-1B*; and *wav2vec2-XLS-R-BN*, a Bengali CTC model trained on standard Bengali.

TABLE VIII

MAIN RESULTS ON THE BANGLAMIX TEST SET (LOWER IS BETTER EXCEPT CS-F1). **BOLD GREEN** MARKS THE BEST OVERALL SYSTEM; “—” DENOTES A METRIC NOT APPLICABLE TO A MONOLINGUAL CTC MODEL.

Model	WER	CER	WER-BN	WER-EN	MIX-ER	WER-CS	CS-F1
<i>Zero-shot baselines</i>							
Whisper-small	1.07	1.02	1.07	1.26	1.15	1.02	0.00
Whisper large-v3	0.91	0.76	0.92	1.08	0.98	0.80	0.14
MMS-1B	0.99	0.57	0.99	1.11	—	0.94	0.04
wav2vec2-XLS-R-BN	0.96	0.62	0.96	1.03	—	0.94	0.04
<i>CTC fine-tuned on BanglaMix</i>							
wav2vec2	0.86	0.49	0.86	1.00	0.97	0.88	0.14
<i>Whisper fine-tuned on BanglaMix (ablation)</i>							
BN_ONLY (A)	0.80	0.49	0.80	1.00	0.94	0.82	0.01
CS_ONLY (B)	0.88	0.55	0.89	0.76	0.82	0.81	0.28
ALL (proposed)	0.80	0.49	0.80	0.93	0.91	0.82	0.19

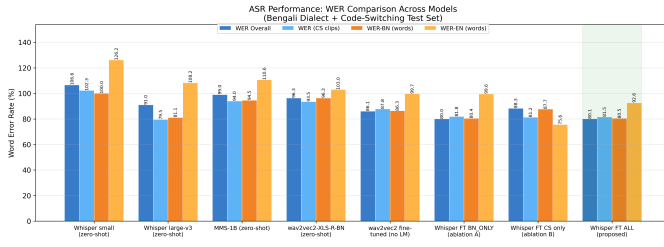


Fig. 4. Word error rate across all models. Fine-tuned systems are clearly separated from zero-shot baselines.

C. Significance Testing

A paired bootstrap [48] ($N=10,000$ resamples over test clips) is used, computing corpus WER on each resample, together with 95% confidence intervals. The paired design is appropriate because all models are evaluated on the identical test set.

VI. RESULTS

Table VIII reports the main results and Fig. 4 visualizes overall WER.

Zero-shot models fail on dialectal CS: Whisper large-v3, despite 680k hours of pre-training, reaches only 0.91 WER, and the same-size Whisper-small is worse (1.07 WER). Multilingual pre-training alone does not transfer to dialectal code-switched Bengali.

Fine-tuning helps substantially: Fine-tuning Whisper-small on BanglaMix lowers WER from 1.07 to 0.80 and CER from 1.02 to 0.49—a 25% relative WER reduction on an identical architecture, which isolates the value of the data from model scale.

The proposed model is strongest under silver-label evaluation: The ALL system attains 0.80 WER / 0.49 CER, significantly better than every zero-shot baseline and the fine-tuned CTC model (0.86) at $p < 0.001$ (paired bootstrap; Section VII)—for example, it improves on zero-shot large-v3 (0.91). On overall WER it is statistically indistinguishable

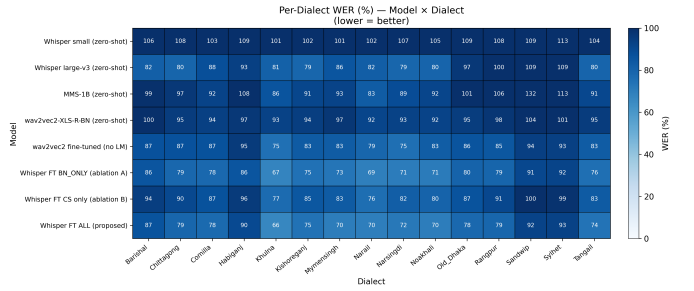


Fig. 5. Model $\times$ dialect WER heatmap. Darker is higher WER.

from the BN-only ablation ($p=0.54$); as the ablation below shows, the value of training on all data lies in code-switch handling rather than in overall WER. These comparisons are against the silver-standard labels; when re-scored against human references on the validated subset the ranking partially reverses (Section VII-I), reflecting label-source bias—which is why we position BanglaMix as a benchmark rather than crowning a single system.

Ablation: BN_ONLY ties ALL on overall WER (0.80; difference not significant, $p=0.54$) but barely recognizes English (CS-F1 0.01); CS_ONLY best preserves English (WER-EN 0.76, CS-F1 0.28) but sacrifices overall accuracy (0.88). Training on ALL clips gives the best balance: overall accuracy on par with the best ablation while substantially improving code-switch handling.

A. Per-Dialect Results

Table IX reports per-dialect WER for the proposed system and Fig. 5 shows the model $\times$ dialect heatmap. The most phonologically distinct dialect, Sylhet (27% of the test set), is the hardest across all models; central dialects (Khulna, Mymensingh, Narail, Noakhali) are easiest.

Table X compares three representative models per dialect. The proposed model is best or tied on 14 of 15 dialects; only on Barishal does the larger zero-shot Whisper large-v3 win,

TABLE IX
PER-DIALECT WER/CER FOR THE PROPOSED WHISPER ALL MODEL.

Dialect	Clips	WER	CER
Sylhet	1,111	0.93	0.65
Old Dhaka	360	0.78	0.53
Sandwip	176	0.92	0.67
Barishal	282	0.87	0.54
Chittagong	198	0.79	0.46
Khulna	120	0.66	0.39
Kishoreganj	743	0.75	0.36
Comilla	157	0.78	0.42
Narail	464	0.70	0.41
Rangpur	165	0.79	0.51
Narsingdi	60	0.72	0.39
Mymensingh	56	0.70	0.35
Habiganj	75	0.90	0.52
Tangail	51	0.74	0.39
Noakhali	70	0.70	0.36
Overall	4,088	0.80	0.49

TABLE X
PER-DIALECT WER FOR REPRESENTATIVE MODELS. **BOLD GREEN** = BEST PER DIALECT.

Dialect	large-v3	wav2vec2	Whisper ALL
Sylhet	1.09	0.93	0.93
Old Dhaka	0.97	0.86	0.78
Sandwip	1.09	0.94	0.92
Barishal	0.82	0.87	0.87
Chittagong	0.80	0.87	0.79
Khulna	0.81	0.75	0.66
Kishoreganj	0.79	0.83	0.75
Comilla	0.88	0.87	0.78
Narail	0.82	0.79	0.70
Rangpur	1.00	0.85	0.79
Narsingdi	0.79	0.75	0.72
Mymensingh	0.86	0.83	0.70
Habiganj	0.93	0.95	0.90
Tangail	0.80	0.83	0.74
Noakhali	0.80	0.83	0.70

indicating that model scale still matters for some dialects and motivating larger fine-tuned backbones as future work.

VII. ANALYSIS AND DISCUSSION

A. Error Types and English-Word Survival

Table XI breaks WER into substitution (S), deletion (D), and insertion (I) rates, and reports English-word survival (EN-Surv), the fraction of English reference words that appear in the hypothesis on CS clips. Fig. 6 visualizes the S/D/I breakdown.

A clear pattern emerges. Monolingual CTC models (MMS, wav2vec2) and the Bengali-only ablation essentially never reproduce English words (EN-Surv at most 0.01): they delete English or map it to Bengali script. Code-switch training raises this sharply—CS_ONLY reaches 0.37, the best among fine-tuned models, and ALL retains 0.13 while keeping the lowest overall WER. Zero-shot Whisper large-v3 preserves English best of all (0.65) but at a far higher overall WER (0.91), as it captures neither dialect phonology nor a coherent transcript. This confirms that sequence-to-sequence models with a shared

TABLE XI
ERROR-TYPE RATES AND ENGLISH-WORD SURVIVAL (EN-SURV).

Model	S	D	I	EN-Surv
Whisper large-v3 (zs)	0.31	0.48	0.12	0.65
Whisper-small (zs)	0.25	0.75	0.07	0.12
MMS-1B	0.66	0.13	0.19	0.00
wav2vec2 (zs)	0.52	0.36	0.08	0.00
wav2vec2 (FT)	0.52	0.23	0.10	0.00
Whisper BN_ONLY	0.45	0.21	0.14	0.01
Whisper CS_ONLY	0.53	0.19	0.16	0.37
Whisper ALL	0.45	0.21	0.14	0.13

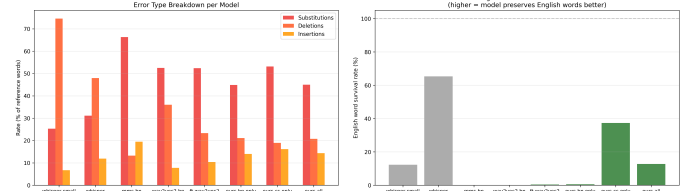


Fig. 6. Substitution / deletion / insertion rates per model.

multilingual vocabulary, fine-tuned on code-switched data, handle CS words far more faithfully than monolingual CTC models.

B. Where Code-Switching Difficulty Lives

We first note the scope of the CS evidence: the per-clip CS metrics (WER-CS, CS-F1) are computed on the 176 code-switched test clips and are therefore *indicative* rather than high-precision. The central CS finding, however, does not depend on that sample size: it is the *qualitative* English-deletion effect—monolingual and Bengali-only models drop English words almost entirely—which is a large, unambiguous effect visible across thousands of English tokens (Table XI), not a marginal difference on a small subset. Consistent with this, a paired bootstrap ($N=10,000$) on the 176 CS clips yields a wide 95% confidence interval for the proposed model’s WER-CS, $[0.76, 0.87]$, that overlaps those of the other Whisper systems (BN_ONLY $[0.78, 0.86]$; zero-shot large-v3 $[0.75, 0.84]$): whole-clip WER-CS does *not* statistically separate models at this sample size, which is precisely why we treat English-word survival—not WER-CS—as the discriminative code-switching measure.

Table XII reports WER on CS versus Bengali-only clips. The whole-clip gap is small (within ± 0.12): because CS clips are predominantly Bengali, code-switching does not greatly change overall clip WER. The difficulty instead lives in the English tokens themselves, as Table XI shows—English survival is near zero for monolingual and BN-only models. Fig. 7 shows that models never trained on CS collapse to near-zero CS-F1, and that training on code-switched data is what recovers code-switch behavior.

Beyond *whether* English words survive, we also measure whether models reproduce the code-switching *structure*—the

TABLE XII
WER ON CS VS. BENGALI-ONLY CLIPS. GAP = WER-CS − WER-BN.

Model	WER-CS	WER-BN	Gap
Whisper large-v3 (zs)	0.80	0.92	−0.12
Whisper-small (zs)	1.02	1.07	−0.05
MMS-1B	0.94	0.99	−0.05
wav2vec2 (FT)	0.88	0.86	+0.02
Whisper CS_ONLY	0.81	0.89	−0.08
Whisper ALL	0.82	0.80	+0.02

TABLE XIII
CODE-SWITCH STRUCTURE PRESERVATION ON CS CLIPS (WHISPER SEQ2SEQ FAMILY): AVERAGE INTRA-UTTERANCE LANGUAGE SWITCHES PRODUCED, AND THE FRACTION OF REFERENCE SWITCHES PRESERVED. REFERENCE CLIPS AVERAGE 3.48 SWITCHES.

Model	Switches/clip	Switch preserved
Reference (human silver)	3.48	—
Whisper CS_ONLY	2.19	47%
Whisper large-v3 (zs)	1.71	44%
Whisper ALL (proposed)	1.32	33%
Whisper BN_ONLY	0.85	20%

number of intra-utterance language switches—computed over roughly 600 reference switch points (more stable than per-clip WER). We report this for the Whisper seq2seq family, for which code-switching is genuinely representable; monolingual CTC models (MMS) are excluded because they cannot reliably emit Latin script (EN-survival 0.00; Table XI), so any apparent “switches” in their output are character-level artifacts rather than real code-switching. To avoid that artifact, a Latin token is counted as English only when it has at least two letters. Reference CS clips contain on average 3.5 switches, yet every Whisper system produces far fewer (Table XIII): models systematically *collapse* code-switching rather than reproduce it. Among the fine-tuned systems the ordering tracks code-switch supervision—CS-only preserves the most switch structure (47%), Bengali-only the least (20%), and the proposed model lies in between (33%)—confirming from a third independent angle that code-switch training restores switching behaviour. Zero-shot large-v3 preserves a comparable fraction (44%), but only by over-generating text: it also shows the highest English survival (Table XI) yet the worst overall WER (0.91), i.e. it keeps switches at the cost of accuracy. The measure counts switch *density* rather than exact positions, so it upper-bounds positional accuracy.

C. Qualitative Error Analysis

Inspecting model outputs reveals three recurring error patterns, illustrated below with romanized examples (Bengali script omitted for readability).

- **Repetition hallucination.** On out-of-distribution clips, zero-shot Whisper enters degenerate loops, e.g. a one-word reference (“*tai*”) expands to forty repetitions of a single token. This pattern accounts for a large share of the zero-

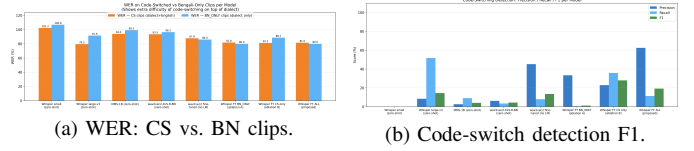


Fig. 7. Code-switching analysis. Models never trained on CS show near-zero CS-F1.

TABLE XIV
ERROR TAXONOMY: DELETION / INSERTION / SUBSTITUTION SHARE OF ALL WORD ERRORS, AND THE FRACTION OF *all* ERRORS THAT ARE BENIGN SPELLING VARIANTS (NEAR-IDENTICAL SUBSTITUTIONS, CHAR-DISTANCE ≤ 0.34).

Model	Del	Ins	Sub	Spelling var.
Whisper large-v3 (zs)	53%	13%	34%	5%
wav2vec2 (FT)	27%	12%	61%	15%
Whisper ALL (proposed)	26%	18%	56%	17%

shot insertion rate and is removed by the repetition-collapse normalization before scoring.

- **English deletion.** Monolingual and Bengali-only models drop English words entirely: a reference “*ami office-e meeting korlam*” is rendered as “*ami ... korlam*”, omitting the English tokens. This is the mechanism behind the near-zero English-word survival in Table XI.
- **Spelling variance.** Fine-tuned models often produce a phonetically correct but differently spelled form (e.g. “*kortasi*” vs. “*kortechi*” for the same spoken word). These count as word errors yet are not genuine recognition failures, which is precisely why CER is much lower than WER and is the more faithful metric for this task.

These patterns explain the headline numbers: fine-tuning chiefly removes hallucination and English deletion, while the residual error is dominated by benign spelling variance over un-standardized dialectal orthography.

To make this quantitative, Table XIV decomposes every word error into deletions, insertions, and substitutions, and further estimates the share of *all* errors that are benign spelling variants—substitutions whose reference and hypothesis words differ by a normalized character distance ≤ 0.34 (i.e. a near-identical spelling). The contrast is stark: zero-shot large-v3 is dominated by *deletions* (53%)—it simply fails to transcribe large spans of dialect speech—whereas fine-tuning halves the deletion share (to $\sim 26\%$) and converts the output into mostly-complete transcripts whose errors are predominantly substitutions, roughly one sixth of which (15–17% of all errors) are mere spelling variants rather than genuine recognition mistakes. This quantifies why CER is substantially lower than WER for the fine-tuned systems and why WER overstates their true error.

D. Ablation Study

Two ablations isolate key design choices. The *training-subset* ablation (Table VIII) shows that BN_ONLY and

TABLE XV
EFFECT OF THE REPETITION-COLLAPSE NORMALIZATION ON THE PROPOSED MODEL (MODEL UNCHANGED; SCORING ONLY).

Configuration	WER	CER
Without repetition collapse	0.96	0.60
With repetition collapse (proposed)	0.80	0.49

CS_ONLY each sacrifice one capability—BN_ONLY collapses on English (CS-F1 0.01) while CS_ONLY loses overall accuracy (0.88 WER)—so only training on all data is strong on both. The *normalization* ablation (Table XV) isolates the repetition-collapse step: applying it lowers the proposed model’s WER from 0.96 to 0.80 and CER from 0.60 to 0.49 *without changing the model*, by preventing Whisper’s degenerate decoding loops from dominating the score.

E. Case Study

A single concrete case illustrates the dominant failure mode. On a short Sylheti test clip whose reference is one word, zero-shot Whisper-small emits the same token roughly forty times in a degenerate loop, producing an enormous per-clip insertion count; the proposed fine-tuned model, decoded with repetition collapse, instead returns a short hypothesis of the correct length. This case explains both why zero-shot insertion rates are high (Table XI) and why the normalization ablation (Table XV) yields such a large apparent gain: a small number of pathological clips, rather than a uniform degradation, drive much of the zero-shot error. The complementary failure on code-switched clips is the deletion of English tokens by monolingual and Bengali-only models, quantified by the near-zero English-word survival in Table XI.

F. Statistical Significance

All WER improvements of the proposed system over the four zero-shot baselines and the fine-tuned CTC model are significant at $p < 0.001$ (paired bootstrap, $N=10,000$). The 95% confidence interval for the proposed system’s WER is [0.79, 0.81], which does not overlap that of any baseline (for example, zero-shot large-v3, [0.90, 0.92]). The proposed model is *not* significantly different from the BN-only ablation on overall WER (0.8009 vs. 0.8004, $p=0.54$); the two are equivalent on this metric, and the benefit of training on all data instead appears in code-switch handling (CS-F1 0.19 vs. 0.01). Effect sizes against the baselines are statistically robust but numerically small (Cohen’s $d = 0.09$ versus large-v3), as expected when comparing models that all operate in a high-error regime on this challenging benchmark.

G. LLM-Ensemble Evaluation

Word error rate captures surface accuracy but not whether a hypothesis is *semantically* faithful or preserves code-switching. As a complementary, reference-free signal, an ensemble of three open LLMs (Mistral-7B [35], LLaMA-3.1-8B [36], Gemma-3-12B [37]) scores each hypothesis from 0–5 on semantic correctness, code-switch preservation, and

TABLE XVI
LLM-ENSEMBLE EVALUATION (MEAN SCORE 0–5; THREE OPEN JUDGES; STRATIFIED 50-CLIP SAMPLE PER MODEL). **BOLD GREEN** = BEST PER COLUMN.

Model	Sem.	CS-pres.	Dialect	Overall
Whisper large-v3	2.41	2.46	2.57	2.48
Whisper CS_ONLY	2.27	2.32	2.05	2.21
Whisper ALL	2.55	2.71	2.43	2.56

dialect naturalness, on a stratified 50-clip sample per model. Table XVI reports the means. The proposed Whisper ALL model attains the highest overall score (2.56/5), the best semantic correctness (2.55), and the best code-switch preservation (2.71); zero-shot large-v3 is marginally higher only on dialect naturalness (2.57 vs. 2.43), and the CS-only ablation is lowest overall (2.21). Inter-annotator (inter-judge) agreement among the three LLM judges—the fraction of judge pairs whose per-clip scores agree within one point on the 0–5 scale—is 0.73–0.85, indicating the ensemble is internally consistent rather than arbitrary. This corroborates the WER findings under an independent, semantics-aware metric—the proposed model is judged best, and its advantage is largest precisely on code-switch preservation—while the mid-range absolute scores reflect the same task difficulty discussed above.

H. Why Absolute Error Rates are High

A WER near 0.80 is high relative to clean ASR (typically 0.05–0.20). Three factors explain this and frame how the numbers should be read. First, the task is intrinsically hard: 15 low-resource dialects combined with code-switching. Second, dialectal Bengali has no standardized spelling, so word-level WER penalizes valid spelling variants—which is why CER (0.49) is the more informative metric here. Third, the reference labels are silver-standard (generated by Whisper large-v3), so part of the measured error reflects label noise rather than model failure. For these reasons BanglaMix is positioned as a *challenging open benchmark* rather than a solved task, and the relative, statistically significant gains are the central result.

I. Evaluation Against Human References

Because the test labels are silver-standard, the main table scores each model against Whisper-generated references, which can favour Whisper-family systems. As a bias-free check, we re-score every model against the *human* transcripts on the 61 human-validated clips (Section III-E); Table XVII reports the result. The silver-label ranking partially reverses: against human references the zero-shot MMS-1B model attains the lowest WER (0.76) and CER (0.41), whereas the proposed Whisper ALL system—best under silver-label scoring on this subset (0.69 WER; cf. 0.80 on the full test set)—rises to 0.83. This is the expected signature of label-source bias: silver labels generated by Whisper inflate the apparent quality of Whisper-family hypotheses, and removing that bias with human references exposes it. Two observations keep the picture coherent. First, fine-tuning still helps *within* the Whisper family even on

TABLE XVII
MODEL ERROR AGAINST HUMAN REFERENCES VS. SILVER LABELS ON
THE 61 HUMAN-VALIDATED TEST CLIPS. WER(HUMAN) IS THE
BIAS-FREE, GOLD-REFERENCE NUMBER; WER(SILVER) IS ON THE SAME
CLIPS FOR COMPARISON. LOWER IS BETTER.

Model	WER _{human}	CER _{human}	WER _{silver}
MMS-1B (zero-shot)	0.76	0.41	0.89
Whisper ALL (proposed)	0.83	0.49	0.69
Whisper BN_ONLY	0.84	0.52	0.70
wav2vec2 (fine-tuned)	0.85	0.48	0.78
wav2vec2-XLS-R-BN (zero-shot)	0.87	0.55	0.92
Whisper CS_ONLY	0.88	0.52	0.81
Whisper large-v3 (zero-shot)	0.90	0.72	0.81
Whisper-small (zero-shot)	1.03	0.99	1.04

human references (Whisper ALL 0.83 vs. zero-shot Whisper-small 1.03 and large-v3 0.90), so the value of training on BanglaMix is real. Second, MMS-1B’s lead is consistent with its much larger (1B) multilingual backbone versus the 244M Whisper-small used here, echoing the per-dialect evidence that backbone scale matters (Section VI). We therefore report results under both reference types and present BanglaMix as an open benchmark rather than crowning a single system. These figures rest on a small sample ($n=61$, 14 code-switched) and are indicative; the *direction* follows directly from the bias mechanism and is unlikely to be an artifact.

VIII. LIMITATIONS

Several limitations frame the scope of this work and should guide interpretation of the results. First, the transcripts are *silver-standard*: they are produced by Whisper large-v3 and validated by an LLM judge rather than by human annotators, so part of the measured error reflects label noise. Our human validation of 61 clips (Section III-E) bounds this noise at WER 0.70 / CER 0.39 and shows it is uneven across dialects (WER 0.26–1.06); because the silver labels originate from Whisper large-v3, the benchmark may also slightly favour Whisper-family systems over other architectures, and a larger human-validated subset remains valuable to tighten these per-dialect estimates. Second, the fine-tuned models use Whisper-small for tractability; larger backbones would likely lower absolute error, as the per-dialect comparison already hints for dialects such as Barishal. Third, code-switched clips are a minority ($\sim 6\%$) of the data, reflecting their natural prevalence; more aggressive query targeting could raise this. Fourth, dialect coverage follows online content availability and is imbalanced, with Sylhet dominating; as broader low-resource ASR surveys note, such imbalance risks systematically underserving speakers of less-represented varieties, so per-dialect reporting and fairness-aware evaluation are important [49]. Finally, the data is drawn from social-media domains (vlogs, comedy, reviews) and may not transfer to formal domains such as broadcast news, healthcare, or education.

IX. REPRODUCIBILITY

To support replication, the dataset (segmented clips, transcripts, and split manifests) and the complete data and eval-

uation pipeline are publicly available (see Data and Code Availability); the fine-tuned model checkpoints will be released upon acceptance. All experiments are driven by a single configuration file that fixes model identifiers, hyperparameters, and the random seed used for the source-level split, so the train/dev/test partition is deterministic. Evaluation applies the identical normalization to every model, and the bootstrap significance test uses a fixed seed, so the reported confidence intervals and p -values are reproducible from the prediction files.

X. CONCLUSION

This work introduced BanglaMix, to our knowledge the first large-scale dataset and benchmark for dialectal Bengali–English code-switching ASR, together with a fully automated, annotator-free construction pipeline. A systematic benchmark of three model families shows that zero-shot multilingual models fail badly on this task, that fine-tuning on BanglaMix yields large and statistically significant gains, and that code-switching difficulty is concentrated in English tokens rather than in overall clip error. The proposed fine-tuned Whisper model reaches 0.80 WER / 0.49 CER and outperforms every zero-shot baseline and the fine-tuned CTC model. Absolute error remains high, reflecting un-standardized dialectal orthography and silver-standard labels, so BanglaMix is released as an open benchmark to drive further progress. Future work includes larger fine-tuned backbones, a human-validated test subset, targeted data collection for the hardest dialects such as Sylhet, and n -gram LM shallow fusion for the CTC models using KenLM [50] and beam-search decoding [51]. Language-model integration and tokenization tuned specifically for dialectal Bengali, following recent low-resource LM-fusion studies [52], [53], is a further promising direction.

DATA AND CODE AVAILABILITY

The BanglaMix dataset (segmented audio clips, silver-standard transcripts, and train/dev/test split manifests) is publicly available on the Hugging Face Hub at https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset, and the complete data-construction and evaluation code is available on GitHub at <https://github.com/niloycste/ASR-Bengali-Local-Dialect->. Both are released for non-commercial research use to support reproducibility; the fine-tuned model checkpoints will be released upon acceptance.

DECLARATION OF COMPETING INTEREST

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] O. Kjartansson, S. Sarin, K. Pipatsrisawat, M. Jansche, and L. Ha, “Crowd-sourced speech corpora for Javanese, Sundanese, Sinhala, Nepali, and Bangladeshi Bengali,” in *Proc. 6th Workshop on Spoken Language Technologies for Under-Resourced Languages (SLTU)*, 2018, pp. 52–55. [Online]. Available: <https://www.openslr.org/37/>

- [2] T. Tashwar Dipto, A. Hossain, R. Sabbir Faruque, M. Rezuwan Hassan, K. Fatema, T. Shome, R. Naswan, M. Foriduzzaman Zihad, M. U. Anam, N. Tasnim *et al.*, “Are ASR foundation models generalized enough to capture features of regional dialects for low-resource languages?” *arXiv e-prints*, 2025, arXiv:2510.
- [3] A. Diwan, R. Vaideeswaran, S. Shah, A. Singh, S. Raghavan, S. Khare, V. Unni, S. Vyas, A. Rajpuria, C. Yarra *et al.*, “MUCS 2021: Multilingual and code-switching ASR challenges for low resource Indian languages,” in *Proc. Interspeech 2021*, 2021.
- [4] M. A. Hasnat, J. Mowla, and M. Khan, “Isolated and continuous Bangla speech recognition: Implementation, performance and application perspective,” in *Proceedings of the International Symposium on Natural Language Processing (SNLP)*, Hanoi, Vietnam, 2007.
- [5] S. T. Swarna, S. Ehsan, M. S. Islam, and M. E. Jannat, “A comprehensive survey on Bengali phoneme recognition,” *arXiv preprint arXiv:1701.08156*, 2017.
- [6] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, “Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks,” in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*. ACM, 2006, pp. 369–376.
- [7] A. Graves, A.-r. Mohamed, and G. Hinton, “Speech recognition with deep recurrent neural networks,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2013, pp. 6645–6649.
- [8] A. Hannun, C. Case, J. Casper, B. Catanzaro, G. Diamos, E. Elsen, R. Prenger, S. Satheesh, S. Sengupta, A. Coates, and A. Y. Ng, “Deep speech: Scaling up end-to-end speech recognition,” *arXiv preprint arXiv:1412.5567*, 2014.
- [9] W. Chan, N. Jaitly, Q. V. Le, and O. Vinyals, “Listen, attend and spell: A neural network for large vocabulary conversational speech recognition,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2016, pp. 4960–4964.
- [10] S. Watanabe, T. Hori, S. Karita, T. Hayashi, J. Nishitoba, Y. Unno, N. E. Y. Soplin, J. Heymann, M. Wiesner, and N. Chen, “ESPnet: End-to-end speech processing toolkit,” in *Interspeech*, 2018, pp. 2207–2211.
- [11] S. Mandal, S. Yadav, and A. K. Rai, “End-to-end Bengali speech recognition,” *arXiv preprint arXiv:2009.09615*, 2020.
- [12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [13] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [14] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [15] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451–3460, 2021.
- [16] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, and F. Wei, “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [17] A. Babu, C. Wang, A. Tjandra, K. Lakhotia, Q. Xu, N. Goyal, K. Singh, P. von Platen, Y. Saraf, J. Pino, A. Baevski, A. Conneau, and M. Auli, “XLS-R: Self-supervised cross-lingual speech representation learning at scale,” in *Proc. Interspeech 2022*, 2022, pp. 2278–2282.
- [18] A. Conneau, A. Baevski, R. Collobert, A. Mohamed, and M. Auli, “Un-supervised cross-lingual representation learning for speech recognition,” in *Interspeech*, 2021, pp. 2426–2430.
- [19] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*. PMLR, 2023, pp. 28492–28518.
- [20] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Xu, A. Baevski, A. Conneau, R. Collobert, Y. Saraf, and M. Auli, “Scaling speech technology to 1,000+ languages,” in *Proceedings of the 41st International Conference on Machine Learning (ICML)*. PMLR, 2024.
- [21] R. Ardila, M. Branson, K. Davis, M. Henretty, M. Kohler, J. Meyer, R. Morais, L. Saunders, F. M. Tyers, and G. Weber, “Common voice: A massively-multilingual speech corpus,” in *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, 2020, pp. 4218–4222.
- [22] S. Biswas, T. Sheikh *et al.*, “A unified denoising and adaptation framework for self-supervised Bengali dialectal ASR,” *arXiv preprint arXiv:2509.00988*, 2025.
- [23] M. S. I. Ridoy, S. Akter, and M. A. Rahman, “Adaptability of ASR models on low-resource language: A comparative study of Whisper and Wav2Vec-BERT on Bangla,” in *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*. IEEE, 2025, pp. 1–6.
- [24] E. Yilmaz, H. van den Heuvel, and D. A. van Leeuwen, “Acoustic and textual data augmentation for improved ASR of code-switching speech,” in *Proc. Interspeech 2018*, 2018, pp. 1933–1937.
- [25] S. Sitaram, K. R. Chandu, S. K. Rallabandi, and A. W. Black, “A survey of code-switched speech and language processing,” *arXiv preprint arXiv:1904.00784*, 2019.
- [26] D.-C. Lyu, T.-P. Tan, E. S. Chng, and H. Li, “SEAME: a Mandarin-English code-switching speech corpus in south-east asia,” in *Interspeech*, 2010, pp. 1986–1989.
- [27] G. I. Winata, S. Cahyawijaya, Z. Liu, Z. Lin, A. Madotto, and P. Fung, “Are multilingual models effective in code-switching?” in *Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching*, 2021, pp. 142–153.
- [28] R. Sennrich, B. Haddow, and A. Birch, “Neural machine translation of rare words with subword units,” in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2016, pp. 1715–1725.
- [29] Y. Yang, Z. Song, J. Zhuo, M. Cui, J. Li, B. Yang, Y. Du, Z. Ma, X. Liu, Z. Wang *et al.*, “GigaSpeech 2: An evolving, large-scale and multi-domain ASR corpus for low-resource languages with automated crawling, transcription and refinement,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025, pp. 2673–2686.
- [30] S. Dhabhi, N. Saleem, S. Bourouis, M. Berrima, and E. Verdú, “End-to-end neural automatic speech recognition system for low resource languages,” *Egyptian Informatics Journal*, vol. 29, p. 100615, 2025.
- [31] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, “Judging LLM-as-a-judge with MT-bench and chatbot arena,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [32] Y. Wang, Z. Yu, Z. Zeng, L. Yang, C. Wang, H. Chen, C. Jiang, R. Xie, J. Wang, X. Xie, W. Ye, S. Zhang, and Y. Zhang, “PandaLM: An automatic evaluation benchmark for LLM instruction tuning optimization,” *arXiv preprint arXiv:2306.05087*, 2023.
- [33] J. Fu, S.-K. Ng, Z. Jiang, and P. Liu, “GPTScore: Evaluate as you desire,” *arXiv preprint arXiv:2302.04166*, 2023.
- [34] Qwen Team, “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024.
- [35] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, and W. El Sayed, “Mistral 7B,” *arXiv preprint arXiv:2310.06825*, 2023.
- [36] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, A. Goyal, A. Hartshorn, A. Yang, A. Mitra, A. Srivankumar *et al.*, “The Llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.
- [37] Gemma Team, M. Riviere, S. Pathak, P. G. Sessa, C. Hardin, S. Bhupatiraju, L. Hussenot, T. Mesnard, B. Shahriari, A. Ramé, J. Ferret, P. Liu, P. Tafti, A. Friesen, M. Casbon, S. Ramos, R. Kumar *et al.*, “Gemma 2: Improving open language models at a practical size,” *arXiv preprint arXiv:2408.00118*, 2024.
- [38] yt-dlp contributors, “yt-dlp: A feature-rich command-line audio/video downloader,” 2021. [Online]. Available: <https://github.com/yt-dlp/yt-dlp>
- [39] Google, “WebRTC voice activity detector,” 2012. [Online]. Available: <https://webrtc.org>
- [40] S. Gandhi, “Fine-tuning whisper for low-resource speech recognition,” HUGGINGFACE Blog, 2023. [Online]. Available: <https://huggingface.co/blog/fine-tune-whisper>
- [41] D. S. Park, W. Chan, Y. Zhang, C.-C. Chiu, B. Zoph, E. D. Cubuk, and Q. V. Le, “SpecAugment: A simple data augmentation method for automatic speech recognition,” in *Interspeech*, 2019, pp. 2613–2617.
- [42] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, “Conformer: Convolution-

- augmented transformer for speech recognition,” in *Interspeech*, 2020, pp. 5036–5040.
- [43] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, and A. M. Rush, “Transformers: State-of-the-art natural language processing,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*. Association for Computational Linguistics, 2020, pp. 38–45.
 - [44] Q. Lhoest, A. V. del Moral, Y. Jernite, A. Thakur, P. von Platen, S. Patil, J. Chaumond, M. Drame, J. Plu, L. Patry, A. McMillan-Major, P. Schmid, S. Gugger, C. Delangue, T. Matuysiak, L. Debut, S. Bekman, P. Cistac, T. Goehringer, V. Mustar, F. Lagunas, A. Rush, and T. Wolf, “Datasets: A community library for natural language processing,” in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*. Association for Computational Linguistics, 2021, pp. 175–184.
 - [45] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Representations (ICLR)*, 2015.
 - [46] A. C. Morris, V. Maier, and P. Green, “jiwer: Evaluate your speech-to-text system with WER, MER, WIL, WIP and CER,” 2004. [Online]. Available: <https://github.com/jitsi/jiwer>
 - [47] HuggingFace, “Evaluate: A HuggingFace library for reproducible evaluation,” 2022. [Online]. Available: <https://github.com/huggingface/evaluate>
 - [48] T. Berg-Kirkpatrick, D. Burkett, and D. Klein, “An empirical investigation of statistical significance in NLP,” in *Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning (EMNLP-CoNLL)*. Association for Computational Linguistics, 2012, pp. 995–1005.
 - [49] S. Hafiz Imam, B. Sani, D. Ketema Gete, B. Yimam Ahamed, I. Said Ahmad, I. Abdulmumin, S. Muhie Yimam, M. Yahuza Bello, and S. H. Muhammad, “Automatic speech recognition for African low-resource languages: Challenges and future directions,” *arXiv e-prints*, 2025, arXiv:2505.
 - [50] K. Heafield, “KenLM: Faster and smaller language model queries,” in *Proceedings of the Sixth Workshop on Statistical Machine Translation (WMT)*. Association for Computational Linguistics, 2011, pp. 187–197.
 - [51] Kensho Technologies, “pyctcdecode: A fast and flexible CTC beam search decoder with KenLM language model support,” 2021. [Online]. Available: <https://github.com/kensho-technologies/pyctcdecode>
 - [52] X. de Zuazo, E. Navas, I. Saratxaga, and I. Hernández Rioja, “Whisper-LM: Improving ASR models with language models for low-resource languages,” *arXiv preprint arXiv:2503.23542*, 2025.
 - [53] Z. Li and J. Niehues, “Enhance contextual learning in ASR for endangered low-resource languages,” in *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC)*, 2025, pp. 1–7.